

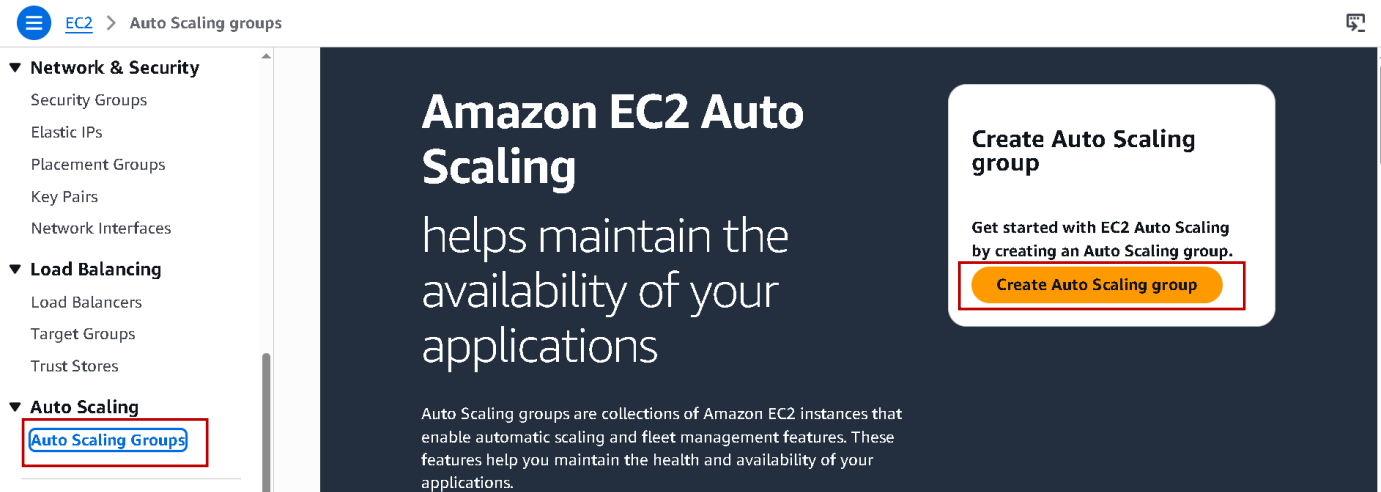
# Dynamic Scaling: -

## Summary: -

I worked on dynamic scaling by automatically increasing or decreasing servers based on CPU usage and traffic load, helping maintain smooth application performance without manual effort.

## Let's Start: -

- Now open your AWS console Search EC2 and open it.



- Now into Nav bar scroll down and click Auto scaling Groups.
- And then click Create Auto scaling group.

## Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

### Name

#### Auto Scaling group name

Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

- Now enter Auto scaling group name.
- Scroll down.

## Launch template [Info](#)

- ❗ For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

### Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

[Create a launch template](#)

- Now we need to create launch Template.
- Understand launch templet and why we need to create with example – assume you have one web application running now during weekdays I have very normal traffic and I have only one EC2 instance which is handling all the traffic, and everything is going well.
- But during Saturday and Sunday you have faced issues because of lots of traffic coming on your web application.
- So, you decided to use Auto scaling and let's increases number of EC2 instances for Weekends.
- Now your requirement is during Saturday and Sunday your EC2 instance will be three and during weekdays only one EC2 instance.
- Now when we use Auto scaling that will add these EC2 instances for sure.
- but what will be the configuration of that EC2 (like I want t2 micro, extra-large instance, this type of operating system, this security group)
- And with launch template we can specify this type of configuration of EC2 on Weekends(Saturday-Sunday).
- Now click Create a launch template.

## Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

### Launch template name and description

Launch template name - *required*

LT-FOR-ASG

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '\*', '@'.

### Template version description

1

Max 255 chars

### Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

- Enter launch template name.
- Enter version.
- And currently off the Auto scaling guidance.
- Scroll down

## ▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

**Recents**

**My AMIs**

**Quick Start**

☒ Don't include in launch template

☐ Recently launched



[Browse more AMIs](#)

Including AMIs from  
AWS, Marketplace and  
the Community

- Now select your AMI click Browse more AMIs.

### Choose an Amazon Machine Image (AMI) [Cancel](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

 Search for an AMI by entering a search term e.g. "Windows"

**Quick Start AMIs (45)**  
Commonly used AMIs

**My AMIs (6)**  
Created by me

**AWS Marketplace AMIs (6082)**  
AWS & trusted third-party AMIs

**Community AMIs (500)**  
Published by anyone

#### Refine results

[Clear all filters](#)

☐ Free tier only [Info](#)

#### ▼ OS category

☐ All Linux/Unix  
☐ All Windows

#### All products (45 filtered, 45 unfiltered)

< 1 >



Amazon Linux  
**Free tier eligible**  
**Verified provider**

Amazon Linux 2023 kernel-6.1 AMI

ami-009be0edec0817ffd (64-bit (x86), uefi-preferred) / ami-080cc69ee28841635 (64-bit (Arm), uefi)

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Platform: amazon

Root device type: ebs

Virtualization: hvm

ENA enabled: Yes

**Select**

☒ 64-bit (x86), uefi-preferred

☐ 64-bit (Arm), uefi

- Now select AMI I am selecting Amazon Linux.
- Simply click Select.

## ▼ Instance type [Info](#) | [Get advice](#)

[Advanced](#)

### Instance type

**t2.micro**

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.017 USD per Hour

On-Demand RHEL base pricing: 0.0268 USD per Hour

On-Demand Linux base pricing: 0.0124 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour

On-Demand SUSE base pricing: 0.0124 USD per Hour

☐ All generations

[Compare instance types](#)

**Additional costs apply for AMIs with pre-installed software**

- Instance type t2 micro.

## ▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

### Key pair name

demo

[Create new key pair](#)

- Now choose Key pair.
- I have a key I use it if you not have simply create it.

## ▼ Network settings [Info](#)

### Subnet [Info](#)

Don't include in launch template

[Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

### Availability Zone [Info](#)

Don't include in launch template

[Enable additional zones](#)

### Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Select existing security group

☒ Create security group

### Security group name - required

Web-SG-231220

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . \_ - / ( ) # , @ [ ] + = & ; { } ! \$ \*

- Now select Create security group.
- And give name of group.

### Description - required [Info](#)

testing

### VPC [Info](#)

vpc-072c4acd761c3b942  
172.31.0.0/16

(default)

- Enter description.

### Inbound Security Group Rules

▼ Security group rule 1 (TCP, 80, 0.0.0.0/0)

Remove

Type | Info

HTTP

Protocol | Info

TCP

Port range | Info

80

Source type | Info

Custom

Source | Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional | Info

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 22, 0.0.0.0/0)

Remove

Type | Info

ssh

Protocol | Info

TCP

Port range | Info

22

Source type | Info

Custom

Source | Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional | Info

e.g. SSH for admin desktop

- Now add two inbound security group.
- 1<sup>st</sup> type HTTP and sources Anywhere (0.0.0.0/0)
- 2<sup>nd</sup> type ssh and Source Anywhere (0.0.0.0/0)

### ▼ Advanced details Info

Domain join directory | Info

Select

↻ Create new directory ↗

IAM instance profile | Info

Select

↻ Create new IAM profile

Hostname type | Info

IP name

DNS Hostname | Info

- So, for that scroll down and open Advanced details.
- And scroll down.

#### User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
</button>
</div>

<div class="footer">
  Powered by <b>Ritesh Behal</b>
</div>

</div>

</body>
</html>
EOF

# Set permissions
chown -R ec2-user:ec2-user /home/ec2-user/
```

☐ User data has already been base64 encoded

#### Virtual server type (instance type)

t2.micro

#### Firewall (security group)

New security group

#### Storage (volumes)

1 volume(s) - 8 GiB

**Free tier:** In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage,

[Cancel](#)

[Create launch template](#)

- Now here paste the script that provided in your resource.
- Now simply click Create launch template.
- Now its ready go back to your Auto scaling Group tab.

## Launch template [Info](#)

**Info** For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

### Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

LT-FOR-ASG

[Create a launch template](#)

### Version

Default (1)

[Create a launch template version](#)

### Description

1

### Launch template

[LT-FOR-ASG](#)

lt-0e002022b82b2ff6e

### Instance type

t2.micro

### AMI ID

ami-009be0edec0817ffd

### Security groups

-

### Request Spot Instances

No

### Key pair name

demo

### Security group IDs

[sg-0a37415d3970732e4](#)

- Now simply click refresh and select created Launch template.
- Now scroll down and click next.

## Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

### VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-072c4acd761c3b942  
172.31.0.0/16 Default



[Create a VPC](#)

### Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets



aps1-az1 (ap-south-1a) | subnet-04ffa8e5e9c79ad3d  
172.31.32.0/20 Default

aps1-az3 (ap-south-1b) | subnet-0bc545e3030c41ef4  
172.31.0.0/20 Default

[Create a subnet](#)

- Now select Availability Zones and subnets because at launch template creation time we cant selected.
- So select ap-south-1a and ap-south-1b
- Now scroll down and simply click next.

## Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

### Select Load balancing options

☒ No load balancer  
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ Attach to an existing load balancer  
Choose from your existing load balancers.

☐ Attach to a new load balancer  
Quickly create a basic load balancer to attach to your Auto Scaling group.

- Now currently select No Load balancer.
- Simply scroll down click next.

## Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

### Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

### Desired capacity

Specify your group size.

1

## Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

### Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity  
1  
Equal or less than desired capacity

Max desired capacity  
5  
Equal or greater than desired capacity

- Now Enter Desired Capacity 1.
- It means I am informing AWS right now I want 1 EC2 instance.

- Min desired capacity 1 and max desired capacity 5.
- Now scroll down and click next.

### Add notifications - *optional* [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

[Add notification](#)

[Cancel](#)

[Skip to review](#)

[Previous](#)

[Next](#)

- Again, simply click next.

### Add tags - *optional* [Info](#)

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

#### Tags (0)

No tags

[Add tag](#)

50 remaining

[Cancel](#)

[Previous](#)

[Next](#)

- Then again click next.

### Step 6: Add tags

[Edit](#)

#### Tags (0)

| Key     | Value | Tag new instances |
|---------|-------|-------------------|
| No tags |       |                   |

No tags

[Preview code](#)

[Cancel](#)

[Previous](#)

[Create Auto Scaling group](#)

- Now simply scroll down and click Create Auto scaling group.
- Now our Auto scaling group is created.
- Now when we go to our EC2 one instance is created because we Enter desired capacity 1 and min capacity 1.
- So, let's check create or not go to EC2 instance.

**Instances (1/7)** Info Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets [Choose filter set](#)

|                                     | Name | Instance ID         | Instance state | Instance type | Status check      | Alarm status                  | Availability Zone |
|-------------------------------------|------|---------------------|----------------|---------------|-------------------|-------------------------------|-------------------|
| <input type="checkbox"/>            |      | i-0145ddd1e123f0e65 | Terminated     | t2.micro      | -                 | <a href="#">View alarms +</a> | ap-south-1b       |
| <input checked="" type="checkbox"/> |      | i-0bdf247fb85730eaa | Running        | t2.micro      | 2/2 checks passed | <a href="#">View alarms +</a> | ap-south-1b       |
| <input type="checkbox"/>            |      | i-00643a25c6eba411a | Terminated     | t2.micro      | -                 | <a href="#">View alarms +</a> | ap-south-1a       |

**i-0bdf247fb85730eaa**

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

#### ▼ Instance summary Info

Instance ID

[i-0bdf247fb85730eaa](#)

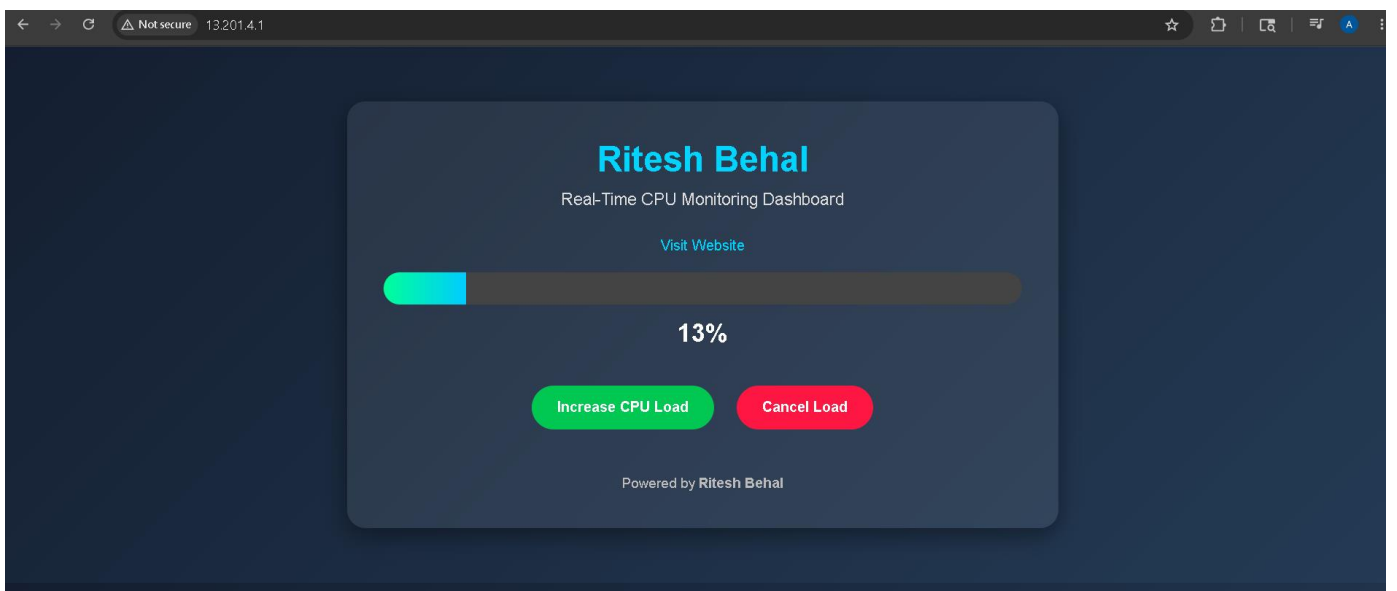
Public IPv4 address

[13.201.4.1](#) | [open address](#)

Private IPv4 addresses

[172.31.9.90](#)

- Now you see instance is created.
- Now select the instance and copy the public IPv4 address and open in new browser tab.



- Now you see its working currently our CPU utilization is 13%.
- Now we go to our Auto Scaling Group tab.
- And open our created ASG.

## ASG-Dynamic-Scaling

**ASG-Dynamic-Scaling Capacity overview** [Edit](#)

[arn:aws:autoscaling:ap-south-1:463646775279:autoScalingGroup:6b3118b5-3d8f-4e77-80ca-e928e170dd88:autoScalingGroupName/ASG-Dynamic-Scaling](#)

|                              |                                |                                                             |                                      |
|------------------------------|--------------------------------|-------------------------------------------------------------|--------------------------------------|
| <b>Desired capacity</b><br>1 | <b>Scaling limits</b><br>1 - 5 | <b>Desired capacity type</b><br>Units (number of instances) | <b>Status</b><br>At desired capacity |
|------------------------------|--------------------------------|-------------------------------------------------------------|--------------------------------------|

**Date created**  
Fri May 15 2026 17:40:26 GMT+0530 (India Standard Time)

[Details](#) [Integrations](#) [Automatic scaling](#) [Instance management](#) [Instance refresh](#) [Activity](#) [Monitoring](#) [Tags](#)

- Now select Automatic Scaling.
- And scroll down.

Dynamic scaling policies (0) [Info](#)
Actions ▾
Create dynamic scaling policy

No dynamic scaling policies have been created

Dynamic scaling policies use real-time data to scale your group based on configurable metrics.

Create dynamic scaling policy

- Now in Dynamic scaling policies click Create dynamic scaling policy.

## Create dynamic scaling policy

**Policy type**

Target tracking scaling
Target tracking scaling ✓
Step scaling
Simple scaling

**Metric type** [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

- Now in policy type we have three options.
- Simple scaling: -
  - SINGLE THRESHOLD
  - THRESHOLD - add 1 instance If CPU Utilization is between 50% and 60%
- Step scaling: -
  - MULTIPLE thresholds
  - Threshold A - add 1 instance when CPU Utilization is between 50% and 60%
  - Threshold B - add 2 instances when CPU Utilization is between 60% and 70%
  - Threshold C - add 3 instances when CPU Utilization is between 70% and 90%
- Target tracking scaling: -
  - It's automatic
  - Set the value and that's Done
  - Self-Optimization.
- Currently I am going with Target tracking scaling.

**Metric type** [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

**Target value**

60

**Instance warmup** [Info](#)

300 seconds

☐ Disable scale in to create only a scale-out policy

Cancel
Create

- Now we choose Metric Type Average CPU utilization.
- Then give target Value 60.
- And simply click Create.

Dynamic scaling policies (1) [Info](#)

Actions ▾

Create dynamic scaling policy

< 1 >

Target Tracking Policy

Policy type

Target tracking scaling

Enabled or disabled

Enabled

Execute policy when

As required to maintain Average CPU utilization at 60

Take the action

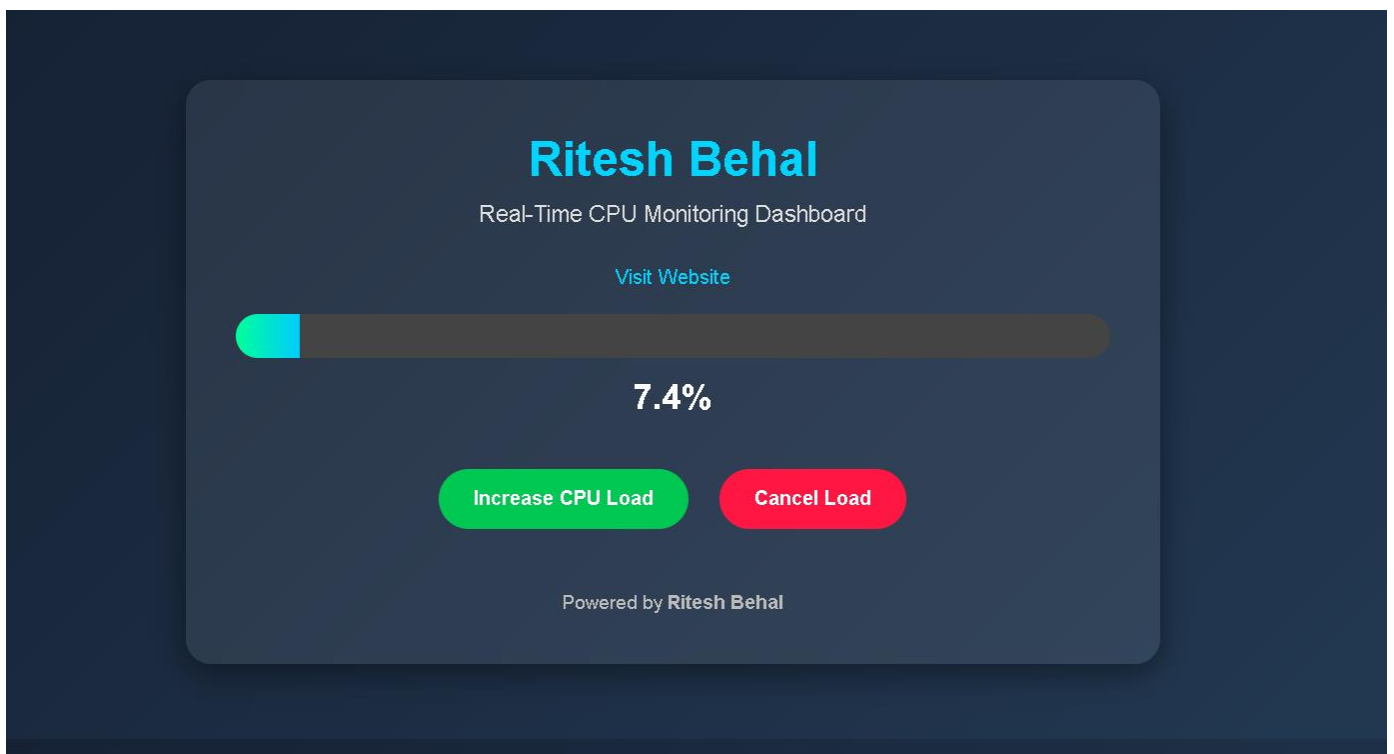
Add or remove capacity units as required

Instances need

300 seconds to warm up before including in metric

☐

- Now see our Policy created.
- It will automatically executed whenever our CPU utilization is above 60%.
- Now go to your website that you opened in another tab.



- Now click on increase CPU Load.

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## Real-Time CPU Monitoring Dashboard

[Visit Website](#)

100%

Increase CPU Load

Cancel Load

Powered by Ritesh Behal

- Now wait and check it dynamically create one more Ec2 instance for you.

### ASG-Dynamic-Scaling

#### ASG-Dynamic-Scaling Capacity overview

Edit

arn:aws:autoscaling:ap-south-1:463646775279:autoScalingGroup:6b3118b5-3d8f-4e77-80ca-e928e170dd88:autoScalingGroupName/ASG-Dynamic-Scaling

Desired capacity  
1

Scaling limits  
1 - 5

Desired capacity type  
Units (number of instances)

Status  
🟢 At desired capacity

Date created  
Fri May 15 2026 17:40:26 GMT+0530 (India Standard Time)

- Now currently you see your Desire Capacity is 1.
- Wait for some minutes and refresh and check.

### ASG-Dynamic-Scaling

#### ASG-Dynamic-Scaling Capacity overview

Edit

arn:aws:autoscaling:ap-south-1:463646775279:autoScalingGroup:6b3118b5-3d8f-4e77-80ca-e928e170dd88:autoScalingGroupName/ASG-Dynamic-Scaling

Desired capacity  
2

Scaling limits  
1 - 5

Desired capacity type  
Units (number of instances)

Status  
🔄 Updating capacity

Date created  
Fri May 15 2026 17:40:26 GMT+0530 (India Standard Time)

- Now you see my desired capacity becomes 2.
- And now check 2<sup>nd</sup> EC2 instance is created or not.
- Go to Ec2 instance.

**Instances (5)** [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets [Choose filter set](#)  < 1 > [Settings](#)

| <input type="checkbox"/> | Name <a href="#">🔗</a> | Instance ID         | Instance state                              | Instance type | Status check      | Alarm status                  | Availability Zone |
|--------------------------|------------------------|---------------------|---------------------------------------------|---------------|-------------------|-------------------------------|-------------------|
| <input type="checkbox"/> |                        | i-003b14685b2673022 | Running <a href="#">🔍</a> <a href="#">🔍</a> | m5.xlarge     | 3/3 checks passed | <a href="#">View alarms +</a> | ap-south-1a       |
| <input type="checkbox"/> |                        | i-06c97099abf45607f | Running <a href="#">🔍</a> <a href="#">🔍</a> | t2.micro      | Initializing      | <a href="#">View alarms +</a> | ap-south-1a       |
| <input type="checkbox"/> |                        | i-04f83bc5838653692 | Running <a href="#">🔍</a> <a href="#">🔍</a> | r8g.xlarge    | 3/3 checks passed | <a href="#">View alarms +</a> | ap-south-1a       |

Select an instance [Settings](#) [▼](#)

- Now you see another one Ec2 is created.
- Now when it becomes active we go to our Opened website tab.

# Ritesh Behal

## Real-Time CPU Monitoring Dashboard

[Visit Website](#)

100%

[Increase CPU Load](#)
[Cancel Load](#)

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- And click on Cancel Load so load from 1<sup>st</sup> Ec2 decreased.
- And if the load is decreased then new created EC2 is automatically Terminated.

**Instances (1/7)** [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets [Choose filter set](#)  < 1 > [Settings](#)

| <input type="checkbox"/>            | Name <a href="#">🔗</a> | Instance ID         | Instance state                                 | Instance type | Status check | Alarm status                  | Availability Zone |
|-------------------------------------|------------------------|---------------------|------------------------------------------------|---------------|--------------|-------------------------------|-------------------|
| <input type="checkbox"/>            |                        | i-05fca4770a13c6b53 | Terminated <a href="#">🔍</a> <a href="#">🔍</a> | m5.xlarge     | –            | <a href="#">View alarms +</a> | ap-south-1a       |
| <input type="checkbox"/>            |                        | i-003b14685b2673022 | Terminated <a href="#">🔍</a> <a href="#">🔍</a> | m5.xlarge     | –            | <a href="#">View alarms +</a> | ap-south-1a       |
| <input checked="" type="checkbox"/> |                        | i-0e697cdb4854b8062 | Terminated <a href="#">🔍</a> <a href="#">🔍</a> | t2.micro      | –            | <a href="#">View alarms +</a> | ap-south-1a       |

- Now see its automatically Terminated.

# ASG-Dynamic-Scaling

ASG-Dynamic-Scaling Capacity overview

Edit

arn:aws:autoscaling:ap-south-1:463646775279:autoScalingGroup:6b3118b5-3d8f-4e77-80ca-e928e170dd88:autoScalingGroupName/ASG-Dynamic-Scaling

|                  |                |                             |                                                                                                       |
|------------------|----------------|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Desired capacity | Scaling limits | Desired capacity type       | Status                                                                                                |
| 1                | 1 - 5          | Units (number of instances) |  Updating capacity |

Date created

Fri May 15 2026 17:40:26 GMT+0530 (India Standard Time)

- And our Desired capacity Becomes 1.

**NOTE:-** If you try to delete EC2 instance from instance page it deleted but we define Desired 1 and min 1 capacity at Auto scaling group creation time so it automatically create 1 EC2 instance so, before sign out from AWS console delete Auto Scaling Group and EC2 created by ASG automatically terminated.